

COS20019 - Cloud Computing Architecture

Week 8: ACA Module 10 Guided Lab - Automating Infrastructure Deployment with AWS CloudFormation

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Due Date: 29/10/2023

Task 1: Deploying a networking layer

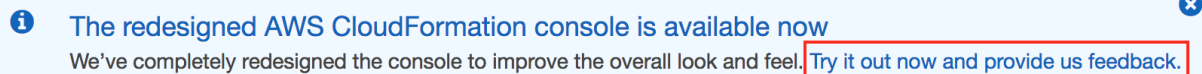
It is a best practice to deploy infrastructure in *layers*. Common layers are:

- Network (Amazon VPC)
- Database
- Application

This way, templates can be reused between systems. For example, you can deploy a common network topology between development, test, and production environments, or deploy a standard database for multiple applications.

In this task, you will deploy an AWS CloudFormation template that creates a *networking layer* by using Amazon VPC.

5. Right-click the following link and download the template to your computer: [lab-network.yaml](#)
If you want, you can open the template in a text editor to see how the AWS resources are defined. Templates can be written in JavaScript Object Notation (JSON) or YAML Ain't Markup Language (YAML). YAML is a markup language that is similar to JSON, but it is easier to read and edit.
6. In the **AWS Management Console**, from the Services menu, choose **CloudFormation**.
7. If you see this message, click [Try it out now and provide us feedback](#):



8. Choose Create stack and configure these settings.

Step 1: Specify template

- **Template source:** Upload a template file
- **Upload a template file:** Click Choose file then select the **lab-network.yaml** file that you downloaded.
- Choose Next

CloudFormation > Stacks > Create stack

Step 1
Create stack

Step 2
Specify stack details

Step 3
Configure stack options

Step 4
Review

Create stack

Prerequisite - Prepare template

Prepare template
Every stack is based on a template. A template is a JSON or YAML file that contains configuration information about the AWS resources you want to include in the stack.

☒ Template is ready ☐ Use a sample template ☐ Create template in Designer

Specify template

A template is a JSON or YAML file that describes your stack's resources and properties.

Template source
Selecting a template generates an Amazon S3 URL where it will be stored.

☐ Amazon S3 URL ☒ Upload a template file

Upload a template file

lab-network.yaml

JSON or YAML formatted file

S3 URL: <https://s3.us-east-1.amazonaws.com/cf-templates-x43x8jn1tvq7-us-east-1/2023-10-29T033552.513Zmff-lab-network.yaml>

9. Step 2: Create Stack

- Stack name: lab-network
- Choose Next

The screenshot shows the AWS CloudFormation console in the 'Specify stack details' step. The left sidebar shows a progress bar with four steps: Step 1 (Create stack), Step 2 (Specify stack details), Step 3 (Configure stack options), and Step 4 (Review lab-network). The main content area is titled 'Specify stack details'. It contains a 'Stack name' section with a text input field containing 'lab-network' and a note that stack names can include letters (A-Z and a-z), numbers (0-9), and dashes (-). Below this is a 'Parameters' section with a note that parameters are defined in the template and allow for custom values. It states 'No parameters' and 'There are no parameters defined in your template'. At the bottom right are 'Cancel', 'Previous', and 'Next' buttons.

10. Step 3: Configure stack options

- In the **Tags** section, enter these values.
 - **Key:** application
 - **Value:** inventory
- Choose Next

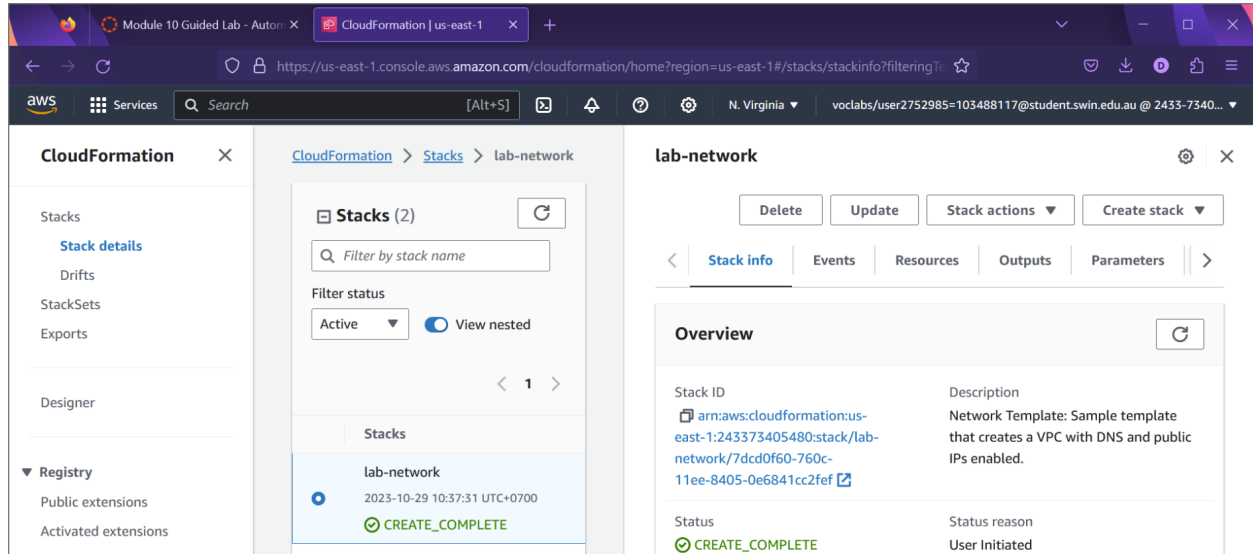
The screenshot shows the AWS CloudFormation console in the 'Configure stack options' step. The left sidebar shows a progress bar with four steps: Step 1 (Create stack), Step 2 (Specify stack details), Step 3 (Configure stack options), and Step 4 (Review lab-network). The main content area is titled 'Configure stack options'. It contains a 'Tags' section with a note that tags are key-value pairs applied to resources. Below this is a table with two columns: 'Key' and 'Value - optional'. The first row has 'application' in the Key column and 'inventory' in the Value column. There is a 'Remove' button next to the value. Below the table is an 'Add new tag' button and a note that you can add 49 more tags. At the bottom right are 'Cancel', 'Previous', and 'Next' buttons.

11. Step 4: Review lab-network

- Choose Create stack
12. The *template* will now be used by AWS CloudFormation to generate a *stack* of resources in the AWS account.
- The specified *tags* are automatically propagated to the resources that are created, which makes it easier to identify resources that are used by particular applications.

Name: Trac Duc Anh Luong - ID: 103488117

13. Choose the **Stack info** tab.
14. Wait for the **Status** to change to **CREATE_COMPLETE**.
Choose **Refresh** every 15 seconds to update the display, if necessary.
You can now examine the resources that were created.



15. Choose the **Resources** tab.
You will see a list of the resources that were created by the template.
If the list is empty, update the list by choosing **Refresh**.

The screenshot displays the AWS CloudFormation console interface. On the left, the 'CloudFormation' sidebar shows navigation options like 'Stacks', 'Stack details', 'Drifts', 'StackSets', 'Exports', 'Designer', 'Registry', and 'Feedback'. The main area is titled 'lab-network' and shows the 'Stacks (2)' list with two stacks, both in 'CREATE_COMPLETE' status. The right pane shows the 'Resources (8)' list for the selected stack, displaying details for various AWS resources.

Logical ID	Physical ID	Type
InternetGateway	igw-00daa95414032710a	AWS::EC2::InternetGateway
PublicRoute	rtb-02c6f80721530e1e3 0.0.0.0/0	AWS::EC2::Route
PublicRouteTable	rtb-02c6f80721530e1e3	AWS::EC2::RouteTable
PublicSubnet	subnet-0577423ab89424524	AWS::EC2::Subnet
PublicSubnetNetworkAclAssociation	aclassoc-0947b8008abc85e33	AWS::EC2::SubnetNetworkAclAssociation
PublicSubnetRouteTableAssociation	rtbassoc-01ad89e3b152755f5	AWS::EC2::SubnetRouteTableAssociation
VPC	vpc-0b7d396a1cfff9d5c	AWS::EC2::VPC
VPCGatewayAttachment	lab-n-VPCGa-QMKPJZVXRTF6	AWS::EC2::VPCGatewayAttachment

16. Choose the **Events** tab and scroll through the events log.

The events log shows (from more recent to less recent) the activities that were performed by AWS CloudFormation. Example events include starting to create a resource and then completing the resource creation. Any errors that were encountered during the creation of the stack will be listed in this tab.

The screenshot displays the AWS CloudFormation console interface. On the left, the navigation pane shows 'Stacks' as the selected section. The main area is divided into two panels. The left panel, titled 'Stacks (2)', lists two stacks: 'lab-network' (created at 2023-10-29 10:37:31 UTC+0700) and 'c92520a206145015122523t1w243373405480' (created at 2023-10-29 10:31:41 UTC+0700). Both stacks show a 'CREATE_COMPLETE' status. The right panel, titled 'lab-network', shows the 'Events' tab. It lists 26 events, all with a 'CREATE_COMPLETE' status. The events include the creation of various resources such as 'PublicRoute', 'PublicSubnetNetworkAclAssociation', 'VPCGatewayAttachment', 'PublicSubnetRouteTableAssociation', 'PublicSubnetRouteTable', 'PublicRouteTable', and 'PublicSubnet'.

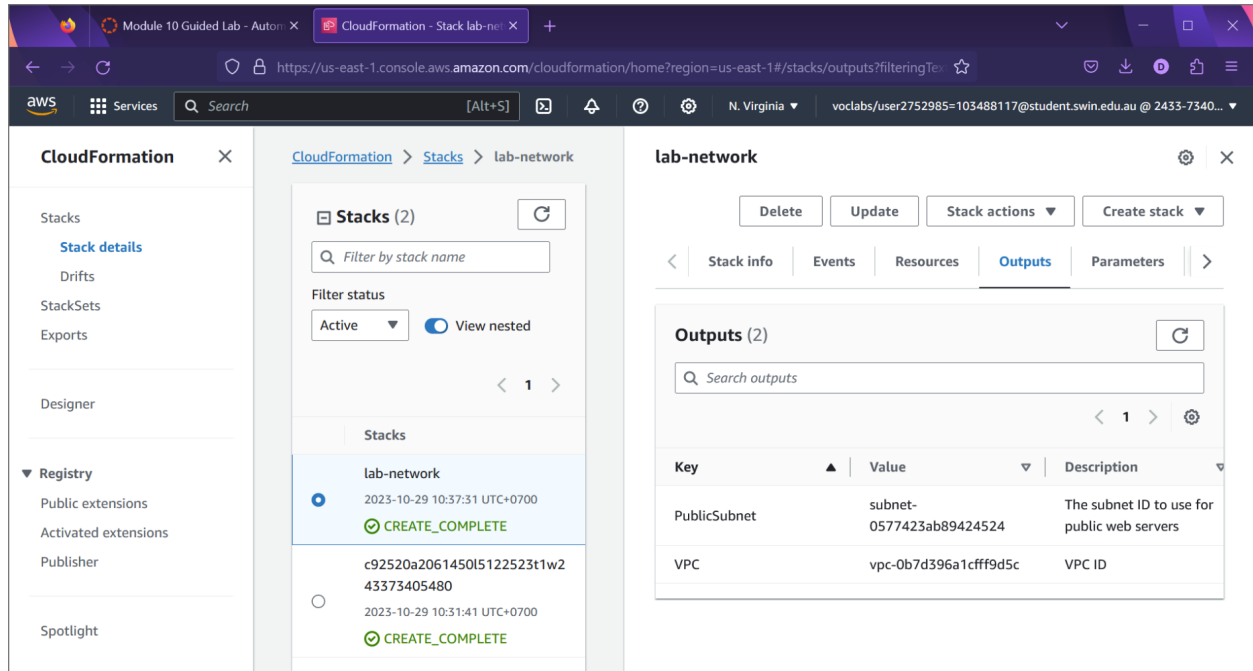
Timestamp	Logical ID	Status
2023-10-29 10:38:11 UTC+0700	lab-network	CREATE_COMPLETE
2023-10-29 10:38:10 UTC+0700	PublicRoute	CREATE_COMPLETE
2023-10-29 10:38:09 UTC+0700	PublicRoute	CREATE_IN_PROGRESS
2023-10-29 10:38:09 UTC+0700	PublicSubnetNetworkAclAssociation	CREATE_COMPLETE
2023-10-29 10:38:08 UTC+0700	PublicRoute	CREATE_IN_PROGRESS
2023-10-29 10:38:07 UTC+0700	VPCGatewayAttachment	CREATE_COMPLETE
2023-10-29 10:38:01 UTC+0700	PublicSubnetRouteTableAssociation	CREATE_COMPLETE
2023-10-29 10:38:01 UTC+0700	PublicSubnetRouteTableAssociation	CREATE_IN_PROGRESS
2023-10-29 10:37:59 UTC+0700	PublicSubnetRouteTableAssociation	CREATE_IN_PROGRESS
2023-10-29 10:37:59 UTC+0700	PublicRouteTable	CREATE_COMPLETE
2023-10-29 10:37:53 UTC+0700	PublicSubnetNetworkAclAssociation	CREATE_IN_PROGRESS
2023-10-29 10:37:53 UTC+0700	PublicSubnetNetworkAclAssociation	CREATE_IN_PROGRESS
2023-10-29 10:37:52 UTC+0700	VPCGatewayAttachment	CREATE_IN_PROGRESS
2023-10-29 10:37:52 UTC+0700	PublicSubnet	CREATE_COMPLETE

17. Choose the **Outputs** tab.

A CloudFormation stack can provide *output information*, such as the ID of specific resources and links to resources.

Two outputs are listed.

- **PublicSubnet:** The ID of the public subnet that was created (for example: subnet-08aafd57f745035f1)
- **VPC:** The ID of the VPC that was created (for example: vpc-08e2b7d1272ee9fb4)



18. Outputs can also be used to provide values to other stacks. This is shown in the **Export name** column. In this case, the VPC and subnet IDs are given export names so that other stacks can retrieve the values. These other stacks can then build resources inside the VPC and subnet that were just created. You will use these values in the next task.

19. Choose the **Template** tab.

This tab shows the template that was used to create the stack—that is, the template that you uploaded while you created the stack. Feel free to examine the template and see the resources that were created. Also feel free to explore the **Outputs** section at the end (this section defined which values to export).

The screenshot displays the AWS CloudFormation console interface. The left sidebar contains navigation options: Stacks, Stack details, Drifts, StackSets, Exports, Designer, Registry, Public extensions, Activated extensions, Publisher, Spotlight, and Feedback. The main content area is titled 'lab-network' and shows the 'Stacks (2)' list. The 'lab-network' stack is highlighted, showing its creation time (2023-10-29 10:37:31 UTC+0700) and status (CREATE_COMPLETE). Below it, another stack is listed with ID 'c92520a206145015122523t1w2' and status 'CREATE_COMPLETE'. The right pane displays the 'Template' for the 'lab-network' stack, which is an AWS::EC2::VPC template. The template includes resources for VPC, Internet Gateway, VPCGatewayAttachment, Public Route Table, PublicRoute, and Public Subnet. The template is in JSON format and includes comments describing the resources.

CloudFormation > Stacks > lab-network

Stacks (2)

Filter by stack name

Filter status: Active View nested

Stacks

Stack	Creation Time	Status
lab-network	2023-10-29 10:37:31 UTC+0700	CREATE_COMPLETE
c92520a206145015122523t1w2	2023-10-29 10:31:41 UTC+0700	CREATE_COMPLETE

lab-network

Delete Update Stack actions Create stack

Resources Outputs Parameters Template Change sets

Template

View in Designer Copy to clipboard

```
AWSTemplateFormatVersion: 2010-09-09
Description: >-
  Network Template: Sample template that creates a VPC with DNS and public IPs
  enabled.

# This template creates:
# VPC
# Internet Gateway
# Public Route Table
# Public Subnet

#####
# Resources section
#####

Resources:

## VPC
VPC:
  Type: AWS::EC2::VPC
  Properties:
    EnableDnsSupport: true
    EnableDnsHostnames: true
    CidrBlock: 10.0.0.0/16

## Internet Gateway
InternetGateway:
  Type: AWS::EC2::InternetGateway

VPCGatewayAttachment:
  Type: AWS::EC2::VPCGatewayAttachment
  Properties:
    VpcId: !Ref VPC
    InternetGatewayId: !Ref InternetGateway

## Public Route Table
PublicRouteTable:
  Type: AWS::EC2::RouteTable
  Properties:
    VpcId: !Ref VPC

PublicRoute:
  Type: AWS::EC2::Route
  DependsOn: VPCGatewayAttachment
  Properties:
    RouteTableId: !Ref PublicRouteTable
    DestinationCidrBlock: 0.0.0.0/0
    GatewayId: !Ref InternetGateway

## Public Subnet
PublicSubnet:
  Type: AWS::EC2::Subnet
  Properties:
    VpcId: !Ref VPC
    CidrBlock: 10.0.0.0/24
```

CloudShell Feedback

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Task 2: Deploying an application layer

Now that you deployed the *network layer*, you will deploy an *application layer* that contains an Amazon Elastic Compute Cloud (Amazon EC2) instance and a security group.

The AWS CloudFormation template will *import* the VPC and subnet IDs from the *Outputs* of the existing CloudFormation stack. It will then use this information to create the security group in the VPC and the EC2 instance in the subnet.

15. Right-click the following link and download the template to your computer: [lab-application.yaml](#)

If you want, you can open the template in a text editor to see how resources are defined.

16. In the left navigation pane, choose **Stacks**.

17. Select **Create stack > With new resources (standard)**, and then configure these settings.

Step 1: Specify template

- **Template source:** Upload a template file
- **Upload a template file:** Click Choose file then select the **lab-application.yaml** file that you downloaded.
- Choose Next

The screenshot shows the AWS CloudFormation console interface for creating a new stack. The left sidebar shows the navigation pane with 'Stacks' selected. The main content area is titled 'Create stack' and shows the 'Specify template' section. The 'Template source' is set to 'Upload a template file'. The 'Upload a template file' section shows the file 'lab-application.yaml' selected. The 'S3 URL' is generated and displayed. The 'Next' button is highlighted in orange.

CloudFormation > Stacks > Create stack

Step 1
Create stack

Step 2
Specify stack details

Step 3
Configure stack options

Step 4
Review

Create stack

Prerequisite - Prepare template

Prepare template
Every stack is based on a template. A template is a JSON or YAML file that contains configuration information about the AWS resources you want to include in the stack.

☒ Template is ready ☐ Use a sample template ☐ Create template in Designer

Specify template

A template is a JSON or YAML file that describes your stack's resources and properties.

Template source
Selecting a template generates an Amazon S3 URL where it will be stored.

☐ Amazon S3 URL ☒ Upload a template file

Upload a template file

lab-application.yaml

JSON or YAML formatted file

S3 URL: <https://s3.us-east-1.amazonaws.com/cf-templates-x43x8jn1tvq7-us-east-1/2023-10-29T034212.828Zply-lab-application.yaml>

Cancel

Step 2: Create Stack

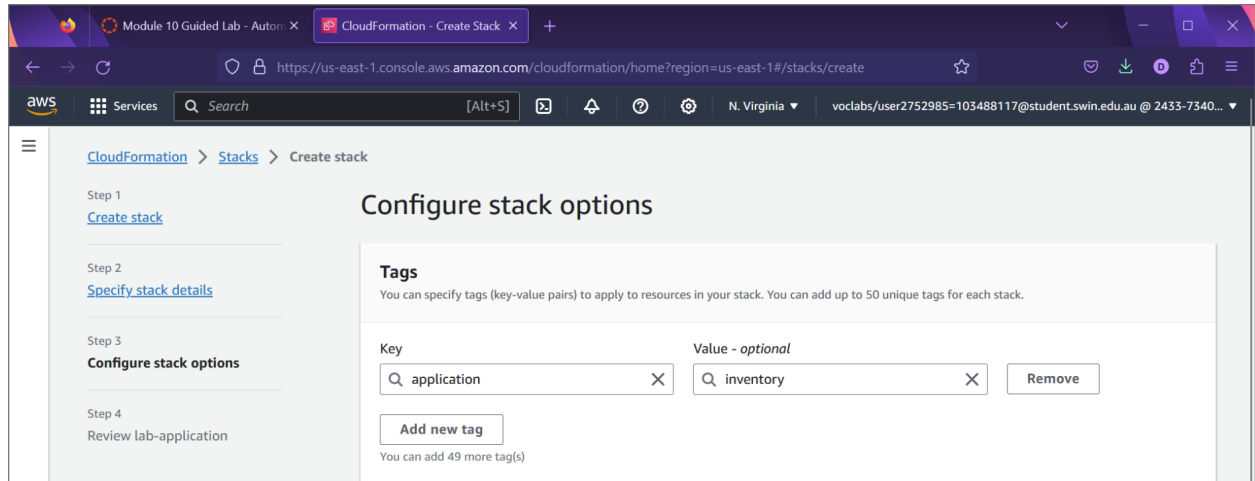
- **Stack name:** lab-application
- **NetworkStackName:** lab-network
- Choose Next

The screenshot shows the AWS CloudFormation console in the 'Specify stack details' step. The left sidebar shows a progress bar with four steps: Step 1 (Create stack), Step 2 (Specify stack details), Step 3 (Configure stack options), and Step 4 (Review lab-application). The main content area is titled 'Specify stack details' and contains two sections: 'Stack name' and 'Parameters'. The 'Stack name' section has a text input field with the value 'lab-application' and a note: 'Stack name can include letters (A-Z and a-z), numbers (0-9), and dashes (-)'. The 'Parameters' section has a note: 'Parameters are defined in your template and allow you to input custom values when you create or update a stack.' It contains two parameter inputs: 'AmazonLinuxAMIID' with the value '/aws/service/ami-amazon-linux-latest/amzn2-ami-hvm-x86_64-gp2' and 'NetworkStackName' with the value 'lab-network'. At the bottom right, there are three buttons: 'Cancel', 'Previous', and 'Next'.

The *Network Stack Name* parameter tells the template the name of the first stack that you created (*lab-network*), so it can retrieve values from the *Outputs*.

Step 3: Configure stack options

- In the **Tags** section, enter these values.
 - **Key:** application
 - **Value:** inventory
- Choose Next

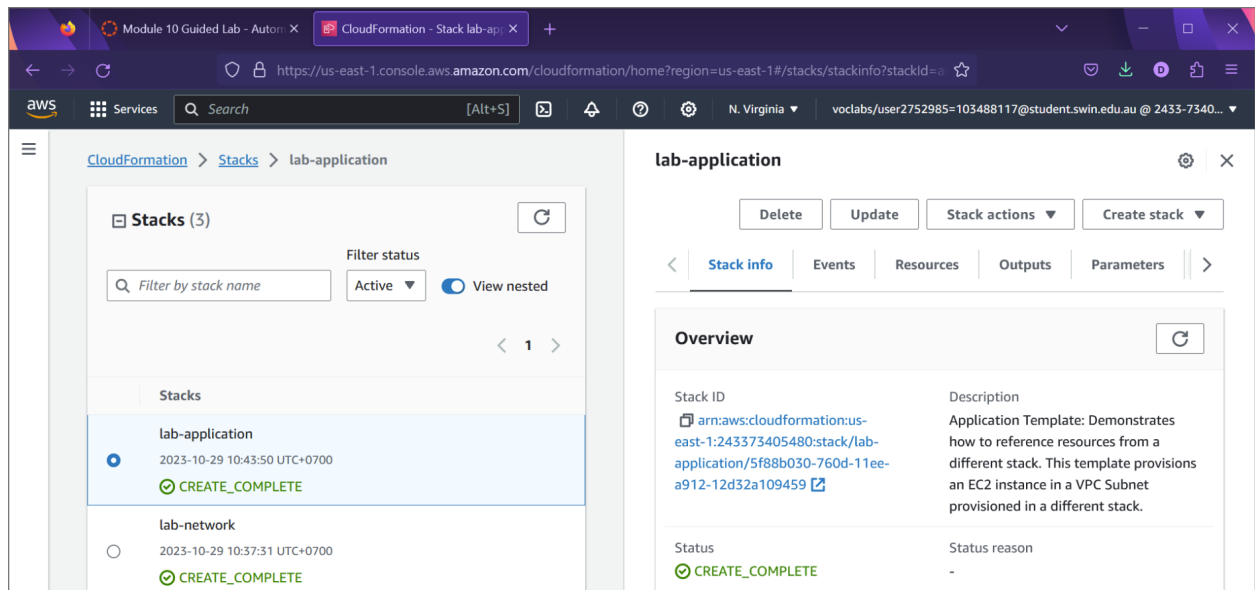


Step 4: Review lab-application

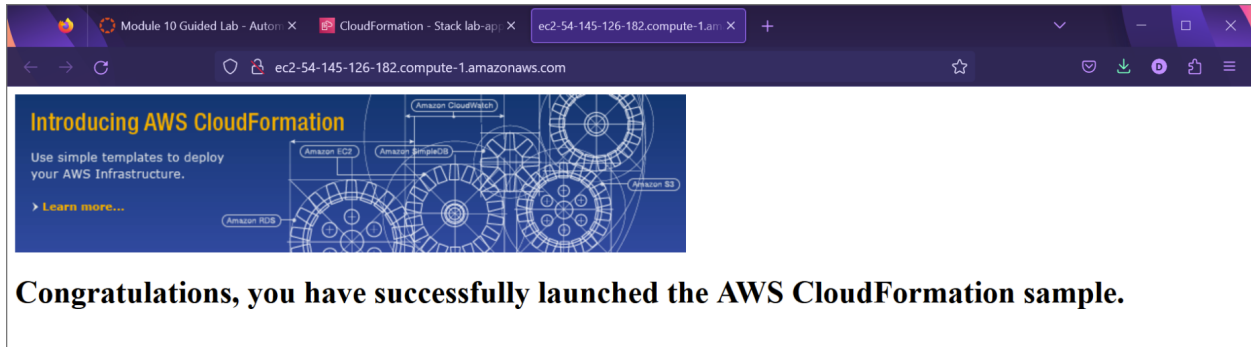
- Choose Create stack

While the stack is being created, examine the details in the **Events** tab and the **Resources** tab. You can monitor the progress of the resource-creation process and the resource status.

18. In the **Stack info** tab, wait for the **Status** to change to **CREATE_COMPLETE**.
Your application is now ready!



19. Choose the **Outputs** tab.
20. Copy the **URL** that is displayed, open a new web browser tab, paste the URL, and press ENTER. The browser tab will open the application, which is running on the web server that this new CloudFormation stack created.



A CloudFormation stack can use reference values from another CloudFormation stack. For example, this portion of the *lab-application* template references the *lab-network* template:

WebServerSecurityGroup:

Type: AWS::EC2::SecurityGroup

Properties:

GroupDescription: Enable HTTP ingress

VpcId:

Fn::ImportValue:

!Sub \${NetworkStackName}-VPCID

The last line uses the *network stack name* that you provided (*lab-network*) when the stack was created. It imports the value of *lab-network-VPCID* from the *Outputs* of the first stack. It then inserts the value into the VPC ID field of the security group definition. The result is that the security group is created in the VPC that was created by the first stack.

Here is another example, which is in the CloudFormation template that you just used to create the application stack. This template code places the EC2 instance into the subnet that was created by the network stack:

SubnetId:

Fn::ImportValue:

!Sub \${NetworkStackName}-SubnetID

It takes the *subnet ID* from the *lab-network* stack and uses it in the *lab-application* stack to launch the instance into the public subnet, which was created by the first stack.

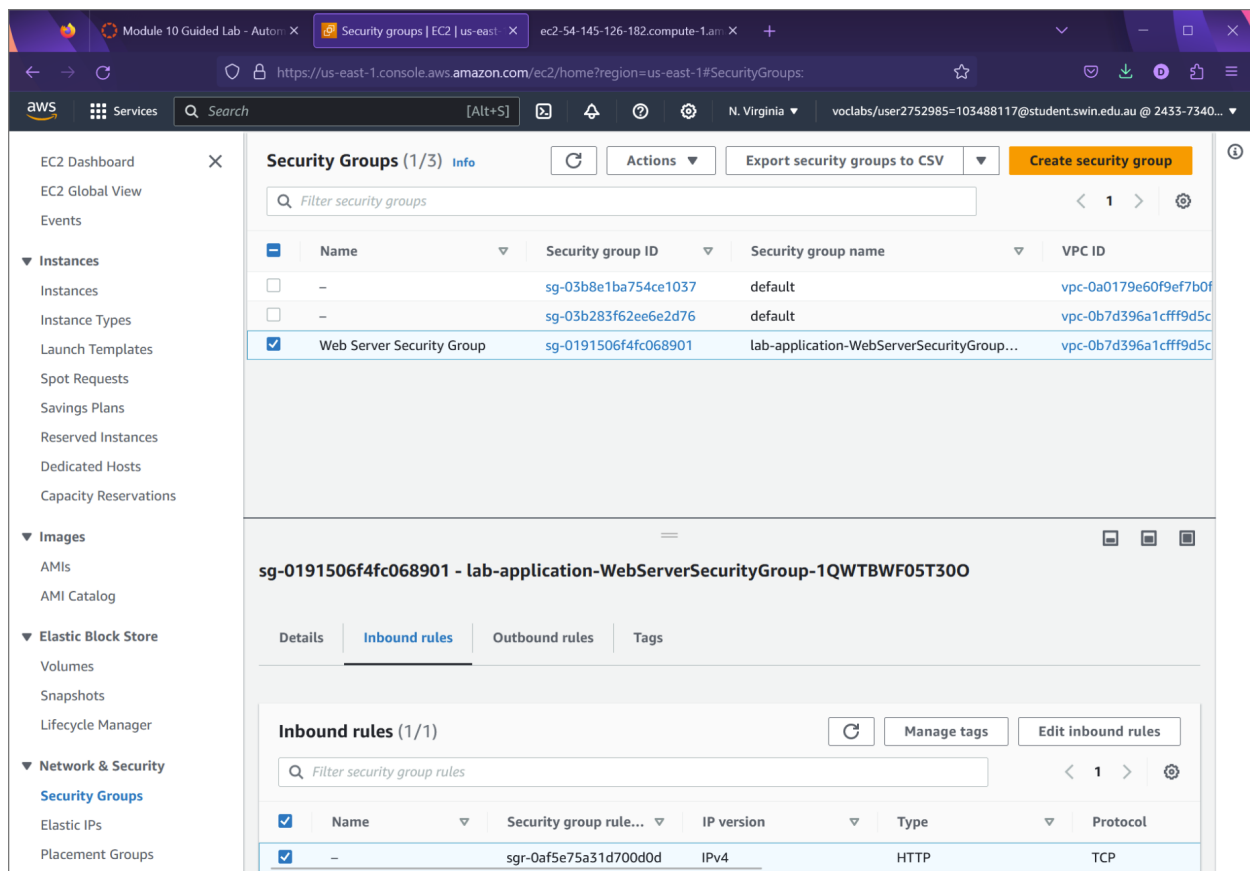
Task 3: Updating a Stack

AWS CloudFormation can also *update* a stack that has been deployed. When you update a stack, AWS CloudFormation will only modify or replace the resources that are being changed. Any resources that are not being changed will be left as-is.

In this task, you will update the *lab-application* stack to modify a setting in the security group.

First, you will examine the current settings for the security group.

21. In the **AWS Management Console**, from the Services menu, choose **EC2**.
22. In the left navigation pane, choose **Security Groups**.
23. Select the check box for **lab-application-WebServerSecurityGroup...**.
24. Choose the **Inbound rules** tab.
Currently, only one rule is in the security group. The rule permits *HTTP* traffic.
You will now return to AWS CloudFormation to update the stack.



25. From the Services menu, choose **CloudFormation**.
26. Right-click the following link and download the updated template to your computer:
[lab-application2.yaml](#)

Name: Trac Duc Anh Luong - ID: 103488117

This template has an additional configuration to permit inbound Secure Shell (SSH) traffic on port 22:

- IpProtocol: tcp
FromPort: 22
ToPort: 22
CidrIp: 0.0.0.0/0

27. In the **Stacks** list of the **AWS CloudFormation console**, select **lab-application**.

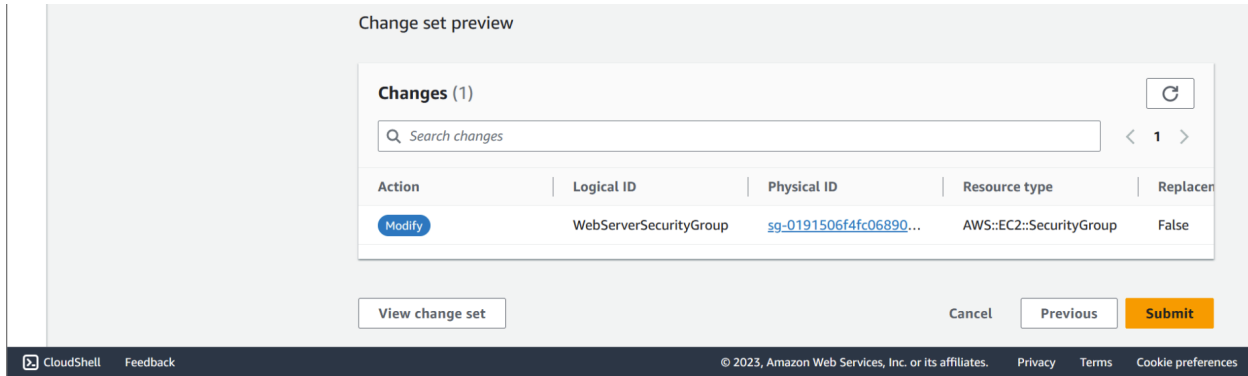
28. Choose Update and configure these settings.

- Select Replace current template
- **Template source:** Upload a template file
- **Upload a template file:** Click Choose file then select the **lab-application2.yaml** file that you downloaded.

The screenshot shows the AWS CloudFormation console interface. The browser address bar indicates the URL: `https://us-east-1.console.aws.amazon.com/cloudformation/home?region=us-east-1#/stacks/update?stackId=arn:aws:cloudformation:us-east-1:103488117:stack/lab-application`. The console header shows the user is logged in as `voclabs/user2752985=103488117@student.swin.edu.au` in the `N. Virginia` region. The breadcrumb navigation is `CloudFormation > Stacks > lab-application > Update stack`. On the left, a sidebar shows the update process steps: **Step 1: Update stack**, **Step 2: Specify stack details**, **Step 3: Configure stack options**, and **Step 4: Review lab-application**. The main content area is titled **Update stack** and contains two sections: **Prerequisite - Prepare template** and **Specify template**. In the **Prepare template** section, the **Replace current template** option is selected. The **Specify template** section shows the **Upload a template file** option selected, with a file named `lab-application2.yaml` chosen. Below this, the **S3 URL** is displayed as `https://s3.us-east-1.amazonaws.com/cf-templates-x43x8jn1tvq7-us-east-1/2023-10-29T035405.842Zbui-lab-application2.yaml`. At the bottom right, there are **Cancel** and **Next** buttons.

29. Choose Next in each of the next *three* screens to advance to the **Review lab-application** page.

In the **Change set preview** section at the bottom of the page, AWS CloudFormation displays the resources that will be updated:



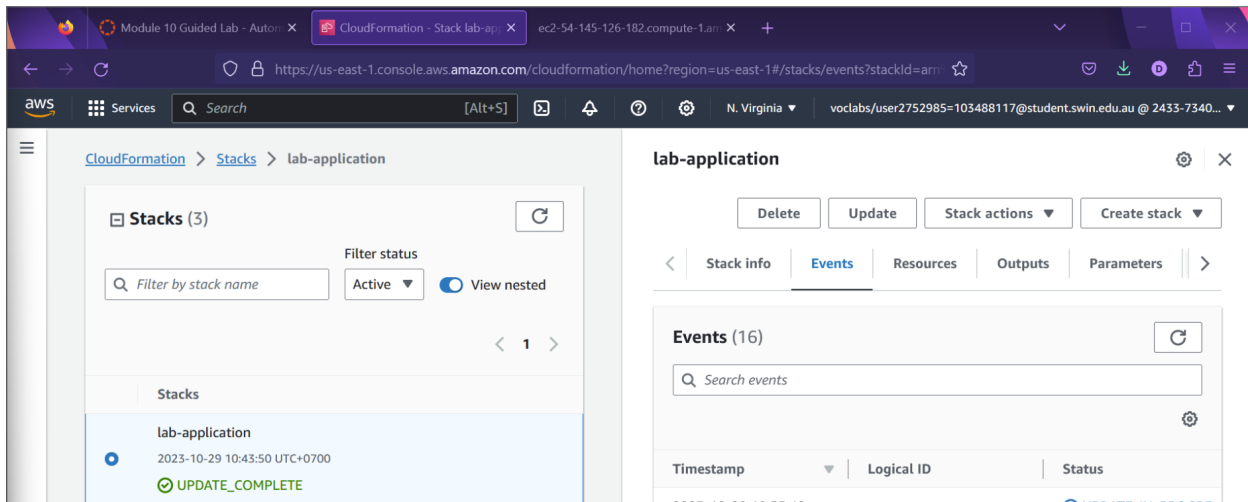
This change set preview indicates that AWS CloudFormation will *Modify* the *WebServerSecurityGroup* without needing to replace it (*Replacement = False*). This change set means that the security group will have a minor change applied to it, and no references to the security group will need to change.

30. Choose Update stack

31. In the **Stack info** tab, wait for the **Status** to change to **UPDATE_COMPLETE**.

Update the status by choosing **Refresh** every 15 seconds, if necessary.

You can now verify the change.



32. Return to the **Amazon EC2 console** and from the left navigation pane, choose **Security Groups**.

33. In the **Security Groups** list, select **lab-application-WebServerSecurityGroup**.

The **Inbound rules** tab should display an additional rule that allows *SSH* traffic over *TCP port 22*.

Name: Trac Duc Anh Luong - ID: 103488117

The screenshot shows the AWS Management Console interface. The left sidebar contains navigation links for EC2 Dashboard, EC2 Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Images, AMIs, AMI Catalog, Elastic Block Store, Volumes, Snapshots, Lifecycle Manager, Network & Security, Security Groups, Elastic IPs, Placement Groups, and Key Pairs. The main content area displays the 'Security Groups (1/3)' page. A table lists three security groups: 'default' (sg-03b8e1ba754ce1037), 'default' (sg-03b283f62ee6e2d76), and 'Web Server Security Group' (sg-0191506f4fc068901). The 'Web Server Security Group' is selected. Below the table, the details for 'sg-0191506f4fc068901 - lab-application-WebServerSecurityGroup-1QWTBWF05T300' are shown. The 'Inbound rules' tab is active, displaying two rules: 'sgr-0f13e3f723b65125e' (SSH, TCP, port 22) and 'sgr-0af5e75a31d700d0d' (HTTP, TCP, port 80).

Name	Security group ID	Security group name	VPC ID	Description
-	sg-03b8e1ba754ce1037	default	vpc-0a0179e60f9ef7b0f	default VPC security group
-	sg-03b283f62ee6e2d76	default	vpc-0b7d396a1cfff9d5c	default VPC security group
Web Server Security Group	sg-0191506f4fc068901	lab-application-WebServerSecurityGroup-1QWTBWF05T300	vpc-0b7d396a1cfff9d5c	Enable HTTP and HTTPS access

Security group rule name	IP version	Type	Protocol	Port range
sgr-0f13e3f723b65125e	IPv4	SSH	TCP	22
sgr-0af5e75a31d700d0d	IPv4	HTTP	TCP	80

This subtask demonstrates how changes can be deployed in a repeatable, documented process. The AWS CloudFormation templates can be stored in a source code repository (such as AWS CodeCommit). This way, you can maintain versions and a history of the templates and the infrastructure that was deployed.

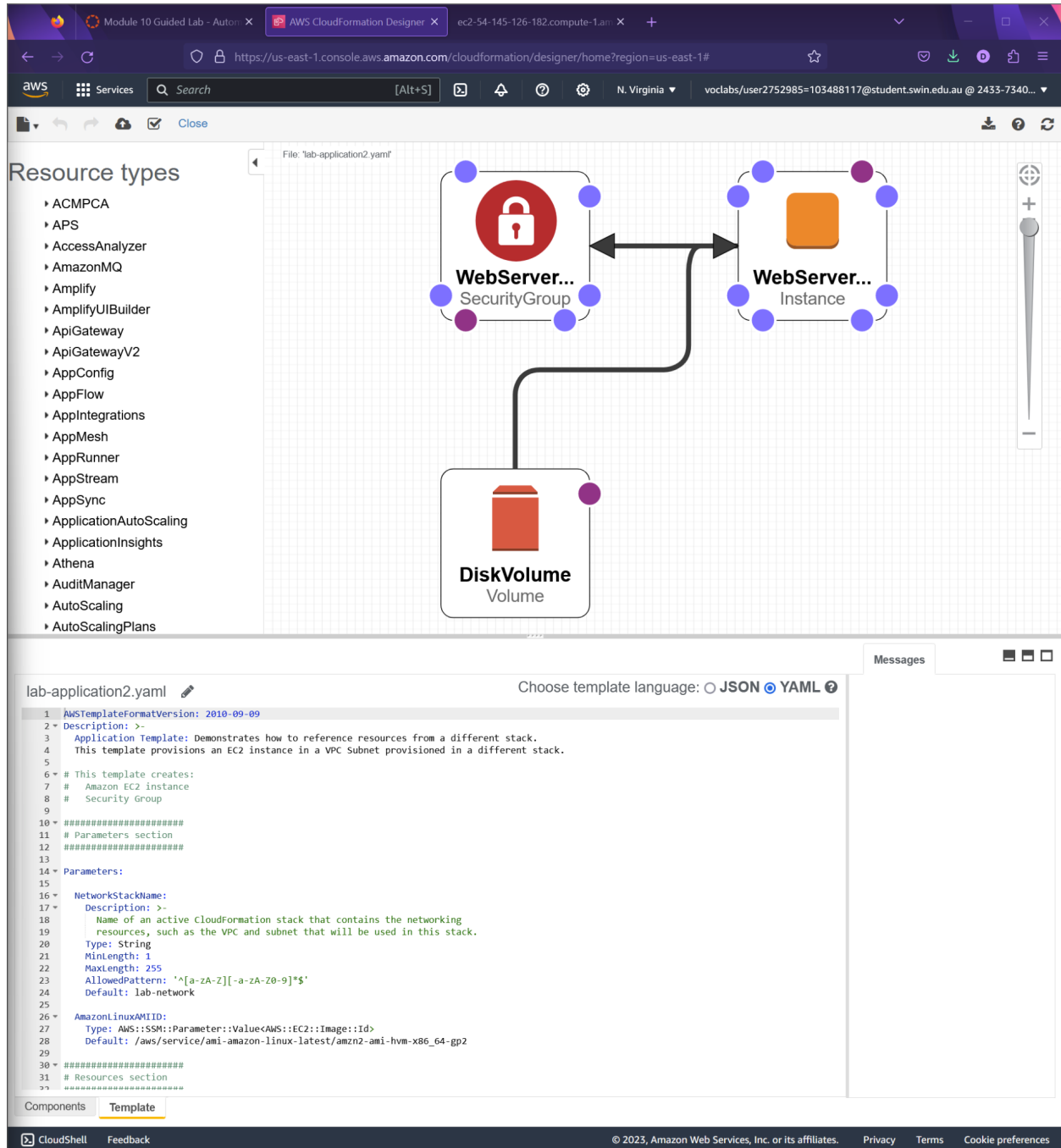
Task 4: Exploring templates with AWS CloudFormation Designer

AWS CloudFormation Designer is a graphic tool for creating, viewing, and modifying AWS CloudFormation templates. With Designer, you can diagram your template resources by using a drag-and-drop interface, and then edit their details through the integrated JSON and YAML editor.

Whether you are a new to AWS CloudFormation or an experienced AWS CloudFormation user, Designer can help you quickly see the interrelationship between a template's resources. It also enables you to easily modify templates.

In this task, you will gain some hands-on experience with Designer.

34. From the Services menu, choose **CloudFormation**.
35. In the left navigation pane, choose **Designer**.
Tip: You might need to expand the left navigation pane by choosing the menu icon.
36. Choose the **File** menu, select **Open > Local file**, and select the **lab-application2.yaml** template that you downloaded previously.
Designer will display a graphical representation of the template:



Instead of drawing a typical architecture diagram, Designer is a visual editor for AWS CloudFormation templates. It draws the resources that are defined in a template and their relationship to each other.

37. Experiment with the features of the Designer. Some things to try are:

- Click the displayed resources. The lower pane will then display the portion of the template that defines the resources.

- Try dragging a new resource—from the **Resource types** pane on the left—into the design area. The definition of the resource will be automatically inserted into the template.
- Try dragging the resource connector circles to create relationships between resources.
- Open the **lab-network.yaml** template that you downloaded earlier in the lab and also explore its resources in Designer.

Task 5: Deleting the stack

When resources are no longer required, AWS CloudFormation can delete the resources built for the stack.

A *deletion policy* can also be specified against resources. It can preserve or (in some cases) back up a resource when its stack is deleted. This feature is useful for retaining databases, disk volumes, or any resource that might be needed after the stack is deleted.

The *lab-application* stack was configured to take a snapshot of an Amazon Elastic Block Store (Amazon EBS) disk volume before it is deleted. The code in the template that accomplishes that configuration is:

DiskVolume:

Type: AWS::EC2::Volume

Properties:

Size: 100

AvailabilityZone: !GetAtt WebServerInstance.AvailabilityZone

Tags:

- Key: Name

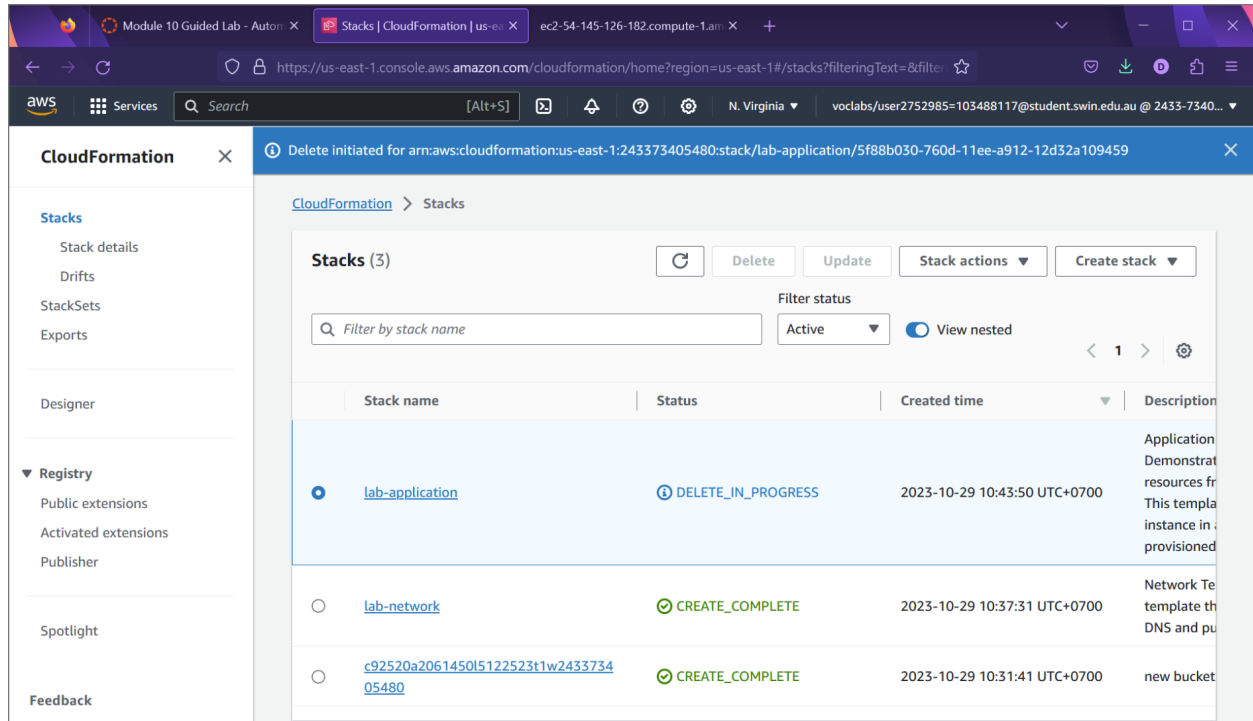
Value: Web Data

DeletionPolicy: Snapshot

The *DeletionPolicy* in the final line directs AWS CloudFormation to create a snapshot of the disk volume before it is deleted.

You will now delete the *lab-application* stack and see the results of this deletion policy.

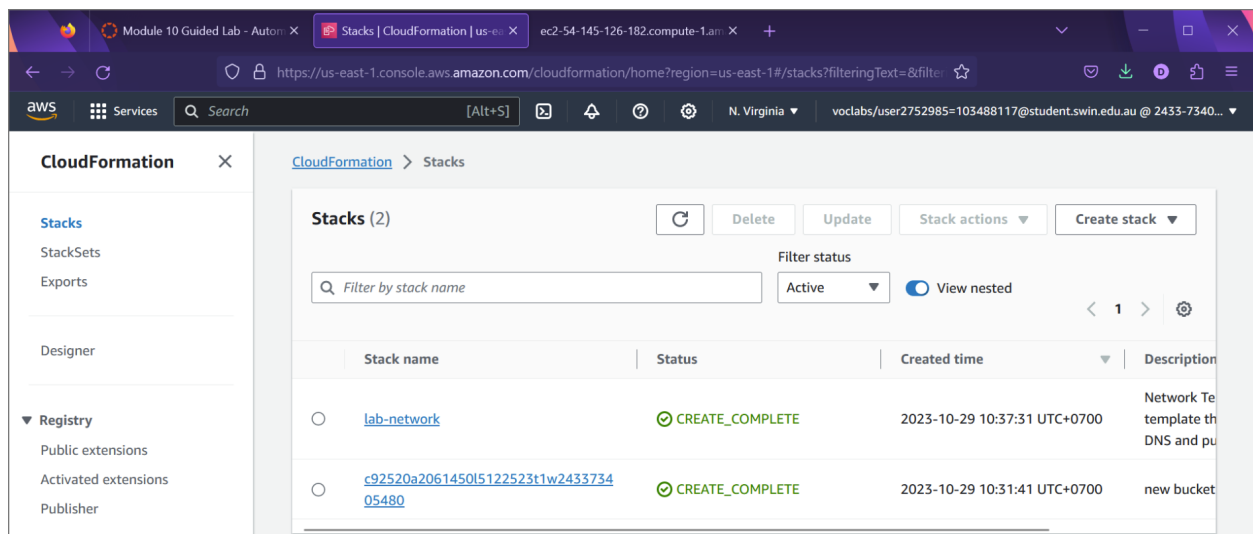
38. Return to the main **AWS CloudFormation console** by choosing the [Close](#) link at the top of the Designer page (choose **Leave page** if prompted).
 39. In the list of stacks, choose the **lab-application** link.
 40. Choose Delete
 41. Choose Delete stack
- You can monitor the deletion process in the **Events** tab and update the screen by choosing **Refresh** occasionally. You might also see an events log entry that indicates that the EBS snapshot is being created.



42. Wait for the stack to be deleted. It will disappear from the stacks list.

The application stack __ removed, but the network stack remained untouched. This scenario reinforces the idea that different teams (for example, the network team or the application team) could manage their own stacks.

You will now verify that a snapshot of the EBS volume was created before the EBS volume was deleted.

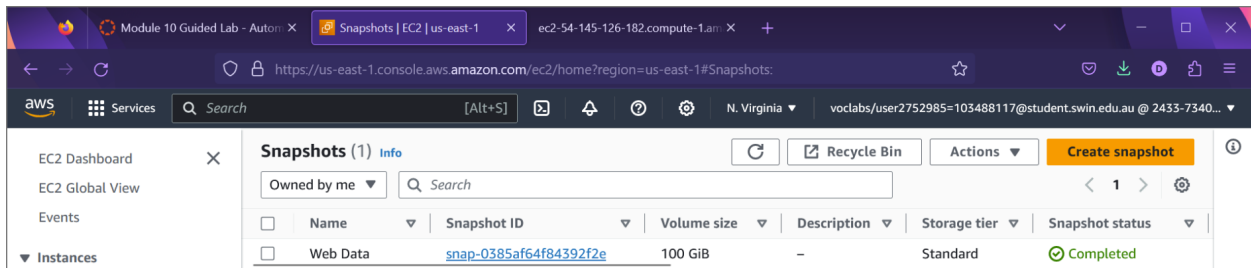


43. From the Services menu, choose **EC2**.

44. In the left navigation pane, choose **Snapshots**.

You should see a snapshot with a **Started** time in the last few minutes.

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Score:

Total score		20/20
[Task 1] Deploy Networking Stack		5/5
[Task 2] Deploy Application Stack		5/5
[Task 3] Update Application Stack		5/5
[Task 5] Delete Application Stack		5/5